

ECE 3337: Signals & Systems Analysis

Lecture 12: The Laplace Transform

**Reference: Chapter 3
(Section 3.4 - 3.7)**

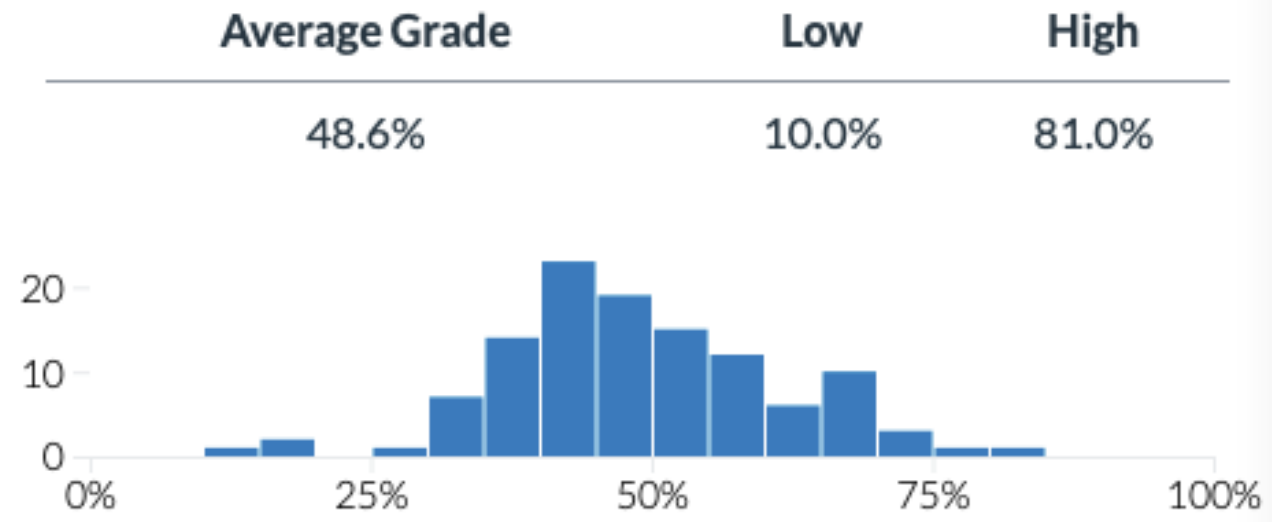
Please turn off your gadgets now



Test #1 has been graded

Test #1

No Due Date





Regrade Requests

- **We have posted the Exam solutions with the detailed grading rubric on Canvas**
- **Compare your answers with the solutions and check if the grades are consistent with the grading rubric**
- **If there is a discrepancy, you may submit a regrade request**
 - **DO NOT write on the exam book**
 - **Write your question / concern on a separate sheet of paper**
 - **Attach this sheet of paper to your exam book, and submit it to your instructor**
 - **You will receive a written response**



Test #1: “Re-Boot” Session

Here is the deal:

- This is for learning only, and not for credit
- The goal is to do better on Test #2. Keep in mind that the remaining chapters build upon the content in Chapters 1 and 2, so we need to get it right
- Study the Test #1 solutions & review Chapters 1 & 2 in advance
- Figure out what you did wrong, and how to correct your mistakes
- Come prepared, just like for a real test, and do your best

Room Cemo Hall Room 109
Friday, March 6, 10AM – 12:00

Goal: Prove to yourself and your study partner(s) that you can ace Test #1 on the re-take

- Get the correct answers
- Do the problems quickly, leaving enough time for checking answers
- Work out best ways to check/verify your answers. Build Confidence.

Math Moment: Completing the Square

A quadratic equation can be rewritten to a more convenient form by Completing the Square, as follows:

$$\begin{aligned}x^2 + bx + c &= x^2 + bx + \left(\frac{b}{2}\right)^2 + c - \left(\frac{b}{2}\right)^2 \\&= \left[x + \frac{b}{2}\right]^2 + c - \left(\frac{b}{2}\right)^2\end{aligned}$$

Example: $x^2 + 4x + 5 = x^2 + 4x + 4 + 5 - 4 = [x + 2]^2 + 1$

Roots: $-2 \pm j$



Laplace Transform Properties

- Using these properties, and the Laplace tables, we can calculate the transforms of common signals
 - Make sure to use the formulas exactly as listed
- We can use the same tables to figure out the inverse Laplace transforms

Table 3-1: Properties of the Laplace transform for causal functions; i.e., $x(t) = 0$ for $t < 0$.

Property	$x(t)$		$\mathbf{X}(s) = \mathcal{L}[x(t)]$
1. Multiplication by constant	$K x(t)$	$\longleftrightarrow$	$K \mathbf{X}(s)$
2. Linearity	$K_1 x_1(t) + K_2 x_2(t)$	$\longleftrightarrow$	$K_1 \mathbf{X}_1(s) + K_2 \mathbf{X}_2(s)$
3. Time scaling	$x(at), \quad a > 0$	$\longleftrightarrow$	$\frac{1}{a} \mathbf{X}\left(\frac{s}{a}\right)$
4. Time shift	$x(t - T) u(t - T)$	$\longleftrightarrow$	$e^{-Ts} \mathbf{X}(s), \quad T \geq 0$
5. Frequency shift	$e^{-at} x(t)$	$\longleftrightarrow$	$\mathbf{X}(s + a)$
6. Time 1st derivative	$x' = \frac{dx}{dt}$	$\longleftrightarrow$	$s \mathbf{X}(s) - x(0^-)$
7. Time 2nd derivative	$x'' = \frac{d^2x}{dt^2}$	$\longleftrightarrow$	$s^2 \mathbf{X}(s) - sx(0^-) - x'(0^-)$
8. Time integral	$\int_{0^-}^t x(t') dt'$	$\longleftrightarrow$	$\frac{1}{s} \mathbf{X}(s)$
9. Frequency derivative	$t x(t)$	$\longleftrightarrow$	$-\frac{d}{ds} \mathbf{X}(s) = -\mathbf{X}'(s)$
10. Frequency integral	$\frac{x(t)}{t}$	$\longleftrightarrow$	$\int_s^\infty \mathbf{X}(s') ds'$
11. Initial value	$x(0^+)$	$=$	$\lim_{s \rightarrow \infty} s \mathbf{X}(s)$
12. Final value	$\lim_{t \rightarrow \infty} x(t) = x(\infty)$	$=$	$\lim_{s \rightarrow 0} s \mathbf{X}(s)$
13. Convolution	$x_1(t) * x_2(t)$	$\longleftrightarrow$	$\mathbf{X}_1(s) \mathbf{X}_2(s)$



Differentiation in the Frequency Domain

$$tx(t) \leftrightarrow -\frac{d\mathbf{X}(s)}{ds}$$

Example: Find the following Laplace transform: $\mathcal{L}[t e^{-at} u(t)]$

$$\mathcal{L}[e^{-at} u(t)] = \frac{1}{s + a}$$

$$\mathcal{L}[t e^{-at} u(t)] = \downarrow -\frac{d}{ds} \left[\frac{1}{s + a} \right] = \frac{1}{(s + a)^2}$$



Integration in the Frequency Domain

$$\frac{x(t)}{t} \leftrightarrow \int_s^{\infty} \mathbf{X}(s') ds'$$

Example: Find the Laplace transform of $x(t) = \frac{\sin(t)}{t} u(t)$

From the table: $\mathcal{L}[\sin(t)u(t)] = \frac{1}{s^2+1}$

$$\begin{aligned} \Rightarrow \mathcal{L}\left[\frac{\sin(t)}{t} u(t)\right] &= \int_s^{\infty} \frac{1}{s'^2 + 1} ds' = \tan^{-1}(s') \Big|_s^{\infty} \\ &= \frac{\pi}{2} - \tan^{-1}(s) \end{aligned}$$



Recap: Combined Example

Calculate the following:

$$\mathcal{L}\left[t^2 e^{-3t} \cos(4t) u(t)\right]$$

When faced with a problem like this, we should first identify the helpful patterns from Tables 3-1 & 3-2:

$\cos(\omega_0 t) u(t)$: Known Laplace transform

e^{-3t} : Frequency Shift

t^2 : Frequency differentiation done twice

Make sure to follow the formulas exactly as stated in the tables



Combined Example

From Table 3-2: Our starting point is

$$\mathcal{L}[\cos(4t)u(t)] = \frac{s}{s^2 + 4^2}$$

Frequency Shift Property: $e^{-at}x(t) \leftrightarrow \mathbf{X}(s + a)$

$$\Rightarrow \mathcal{L}[e^{-3t} \cos(4t)u(t)] = \frac{s + 3}{(s + 3)^2 + 4^2}$$



Combined Example

$$\mathcal{L}\left[e^{-3t} \cos(4t)u(t)\right] = \frac{s+3}{(s+3)^2 + 4^2}$$

Frequency Differentiation: $\mathcal{L}[t x(t)] = -\frac{d}{ds}[\mathbf{X}(s)]$

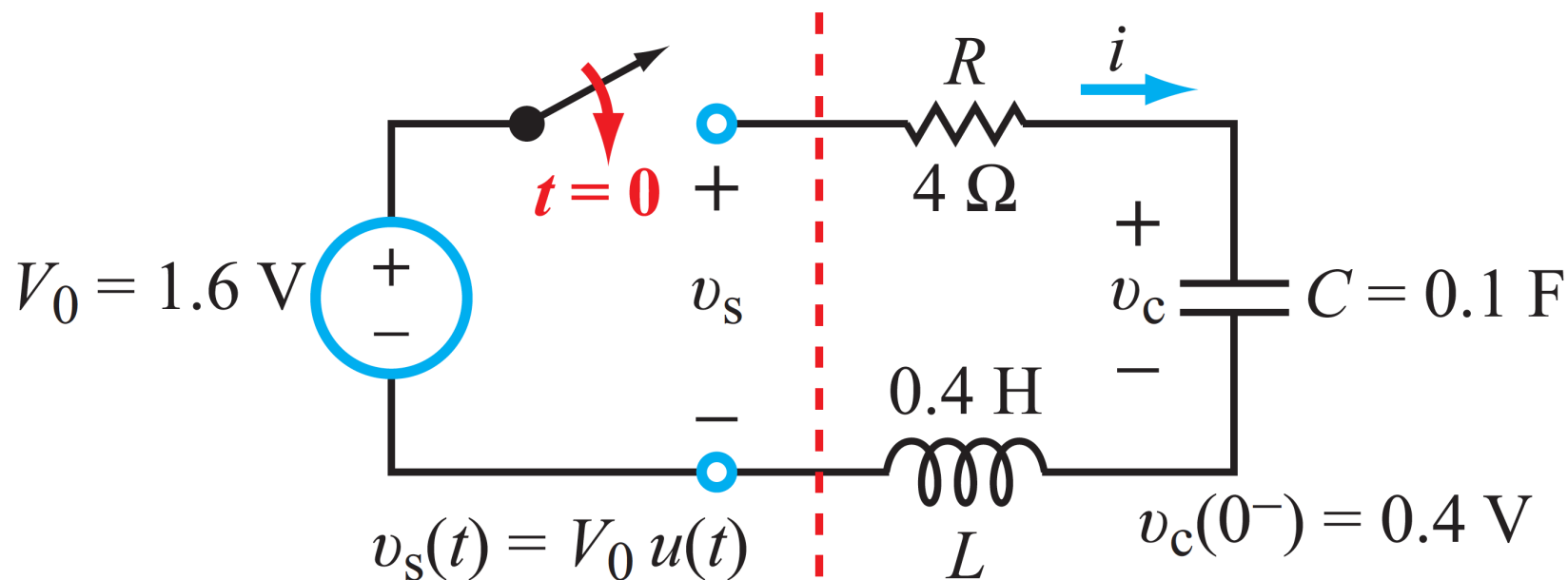
Done Twice: $\mathcal{L}[t^2 x(t)] = \frac{d^2}{ds^2}[\mathbf{X}(s)]$

$$\Rightarrow \mathcal{L}\left[t^2 e^{-3t} \cos(4t)u(t)\right] = \frac{d^2}{ds^2} \left\{ \frac{s+3}{(s+3)^2 + 4^2} \right\}$$



Analyzing Circuits using Laplace Transforms

Find the current $i(t)$ for the circuit below:



$$v_L(t) = L \frac{di(t)}{dt}$$

$$i(t) = C \frac{dv_c(t)}{dt}$$

$$v_c(t) = \frac{1}{C} \int_{0-}^t i(t') dt' + v_c(0-)$$

KVL: $Ri(t) + v_c(t) + v_L(t) = V_0 u(t)$



Entering the Laplace Domain

$$R i(t) + \left[\frac{1}{C} \int_{0^-}^t i(t') dt' + v_C(0^-) \right] + L \frac{di(t)}{dt} = V_o u(t)$$


TRANSFORM EACH TERM

$$\mathcal{L}[R i(t)] = R \mathbf{I}(s)$$

$$\mathcal{L}\left[\frac{1}{C} \int_{0^-}^t i(t') dt'\right] = \frac{1}{C} \frac{\mathbf{I}(s)}{s}$$

$$\mathcal{L}\left[L \frac{di(t)}{dt}\right] = L[\mathbf{sI}(s) - i(0^-)] = L\mathbf{sI}(s)$$

$$\mathcal{L}[V_o u(t)] = V_o \frac{1}{s}$$

$$\mathcal{L}[v_C(0^-)] = \int_{0^-}^{\infty} v_C(0^-) e^{-st} dt = v_C(0^-) \int_{0^-}^{\infty} e^{-st} dt = \frac{v_C(0^-)}{s}$$



The Laplace Domain Approach

$$R i(t) + \left[\frac{1}{C} \int_{0^-}^t i(t') dt' + v_C(0^-) \right] + L \frac{di(t)}{dt} = V_o u(t)$$

FORWARD TRANSFORM

$$R \mathbf{I}(s) + \frac{\mathbf{I}(s)}{Cs} + \frac{v_C(0^-)}{s} + Ls \mathbf{I}(s) = \frac{V_o}{s}$$

SOLVE FOR $\mathbf{I}(s)$

$$\mathbf{I}(s) = \frac{V_o - v_C(0^-)}{L \left[s^2 + \frac{R}{L} s + \frac{1}{LC} \right]} = \frac{3}{(s + 5)^2}$$

INVERSE TRANSFORM

$$i(t) = 3te^{-5t} u(t)$$

VOILA !!



More Complex Example

Consider the following example:

$$X(s) = \underbrace{\frac{4}{s+2} + \frac{6}{(s+5)^2}}_{\text{blue bracket}} + \underbrace{\frac{8}{s^2 + 4s + 5}}_{\text{green bracket}}$$

- Inverse Laplace transforms for the first two terms are listed in Table 3-2 already (easy)
- How do we deal with the third term? The closest entry in the table is:

13

$$e^{-at} \sin(\omega_0 t) u(t) \quad \longleftrightarrow \quad \frac{\omega_0}{(s+a)^2 + \omega_0^2}$$



More Complex Example

By completing the square, we can rewrite this term as follows:

$$\frac{8}{s^2 + 4s + 5} = \frac{8}{s^2 + 4s + 4 + 1} = \frac{8}{(s + 2)^2 + 1}$$

13

$$e^{-at} \sin(\omega_0 t) u(t) \longleftrightarrow \frac{\omega_0}{(s + a)^2 + \omega_0^2}$$

$$\Rightarrow \mathcal{L}^{-1} \left[\frac{8}{(s + 2)^2 + 1} \right] = 8e^{-2t} \sin(t)u(t)$$



Overall Response

$$X(s) = \frac{4}{s+2} + \frac{6}{(s+5)^2} + \frac{8}{(s+2)^2 + 1}$$

$$\Rightarrow x(t) = \left\{ 4e^{-2t} + 6te^{-5t} + 8e^{-2t} \sin(t) \right\} u(t)$$

Practical Observations:

- We have 3 real-valued poles ($-2, -5, -5$), and one pair of complex conjugate poles $\{-2 \pm j\}$
- Real poles produce exponential responses
- Complex conjugate poles produce oscillatory responses



Overall Response

$$\mathbf{X(s)} = \frac{4}{s+2} + \frac{6}{(s+5)^2} + \frac{8}{(s+2)^2+1}$$

$$\Rightarrow x(t) = \left\{ 4e^{-2t} + 6te^{-5t} + 8e^{-2t} \sin(t) \right\} u(t)$$

More Observations:

- The real-valued poles in this case are in the **Left Half Plane (LHP)**, and the corresponding responses are **damped** (they fall off rapidly).
- If the poles were in the **Right Half Plane**, then we would end up with **growing** responses (**unstable**)

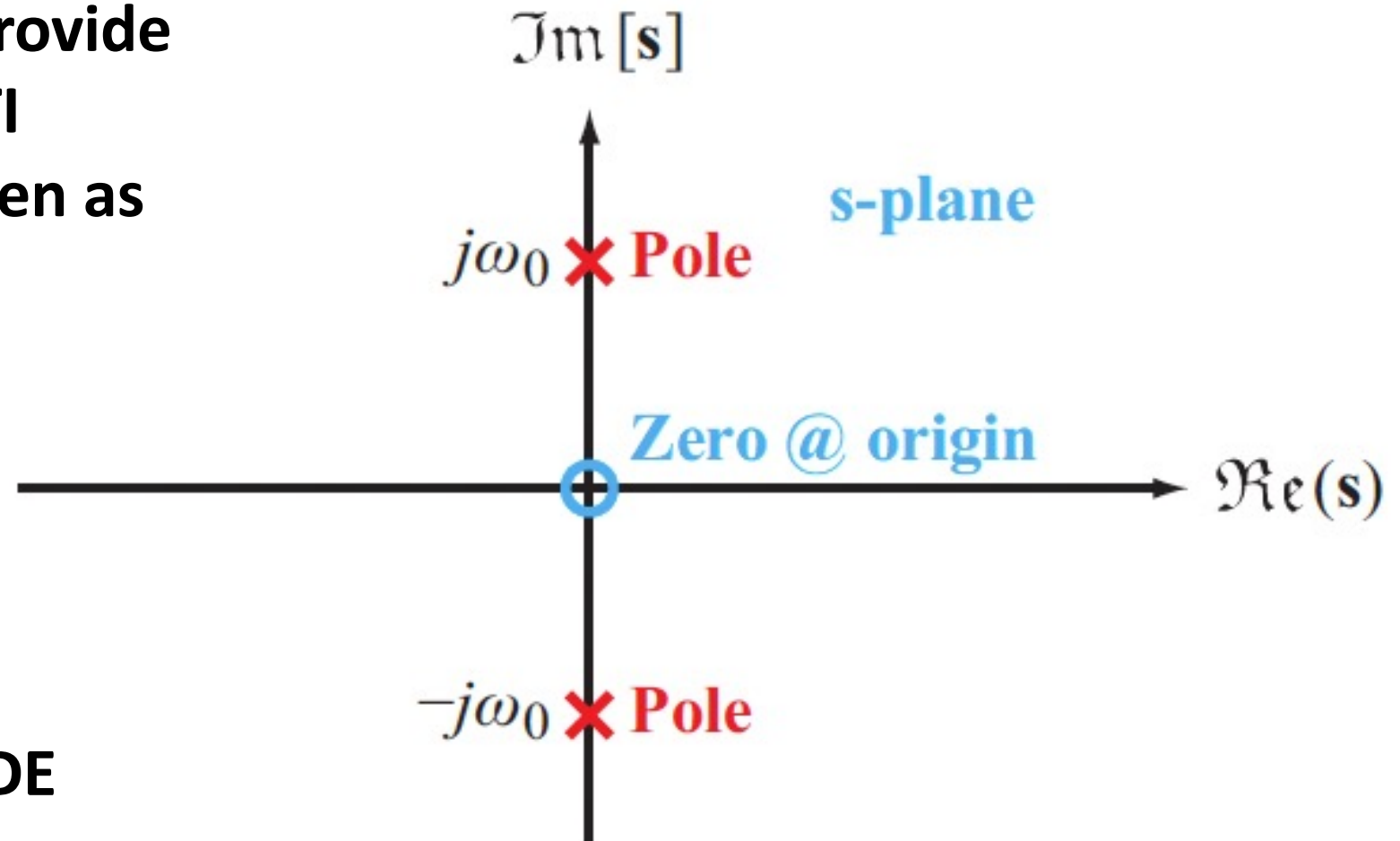


Recap: Pole – Zero Plots

The poles and zeros together provide a complete description of an LTI system/signal that can be written as a ratio of polynomials:

$$X(s) = \frac{N(s)}{D(s)}$$

How does this relate to an LCCDE description of an LTI system?



We can visually interpret the system's behavior from a pole-zero plot!



But What About the $(j\omega)$ Axis?

When we spoke of the “Left Half Plane”, we omitted an important detail

- *Do we include or exclude the vertical axis $\sigma = 0$?*
- When we have $1/s$, i.e., a pole at $s = 0$, it corresponds to a step response $u(t)$ for an impulse input (infinite energy)
- Now consider what happens when we apply a step input $1/s$ instead
 - The output $1/s^2$ is a ramp function that grows indefinitely!

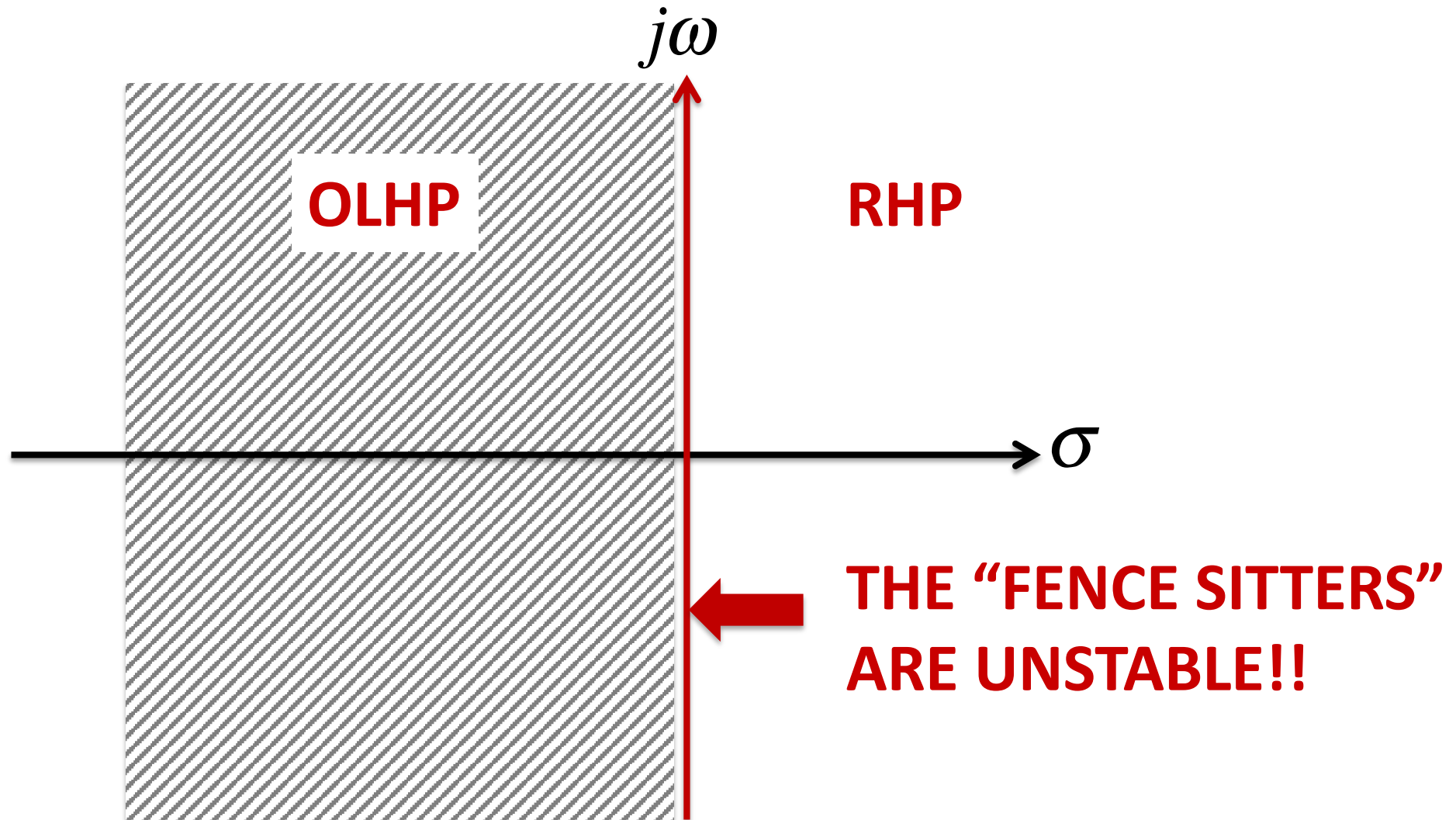
NOT STABLE!

Bottom line: The system is only stable when the poles are in the Open Left Half Plane (OLHP) that excludes the vertical axis



But What About the Axis?

The Open Left Half Plane (OLHP) extends from 0 – to negative infinity





Effects of System Zeros?

Zeros of a system have no bearing on stability since they do not lead to infinite responses

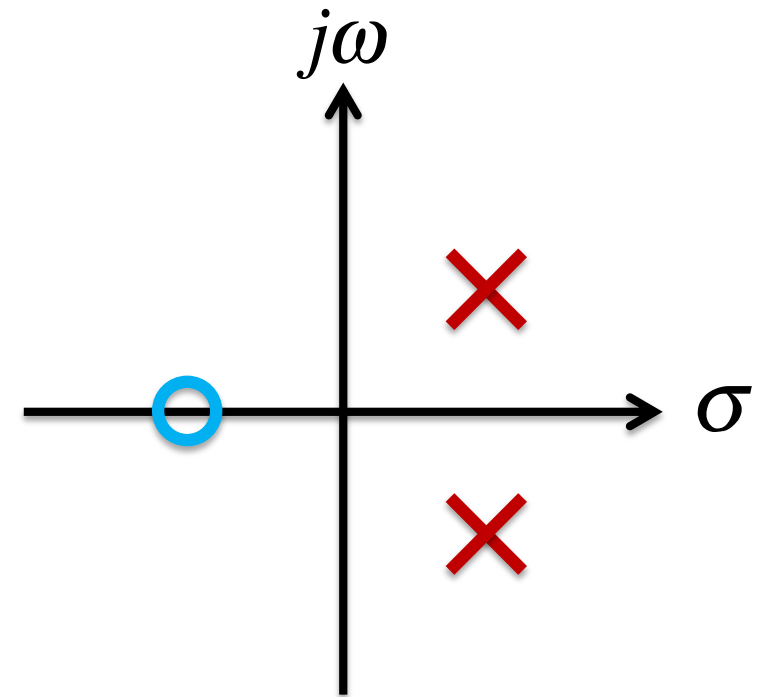
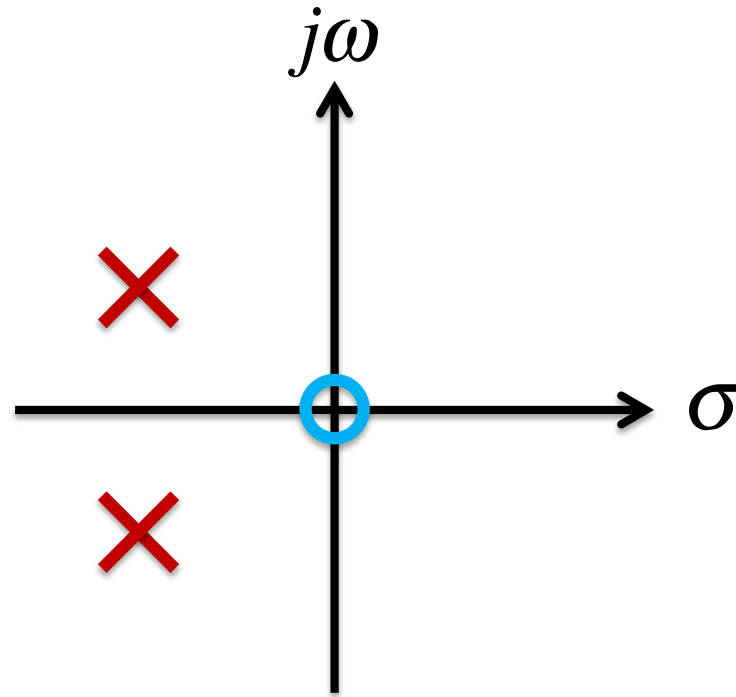
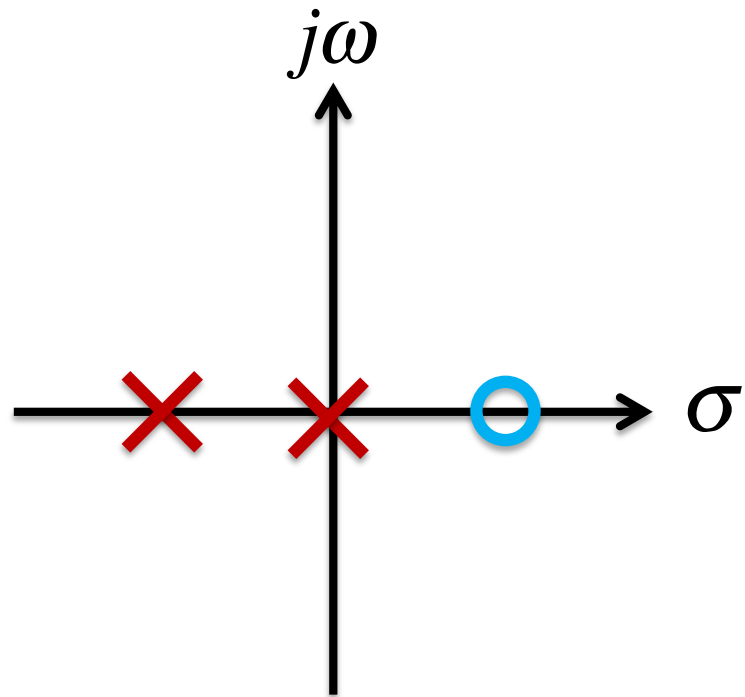
- Does not matter where they are located (LHP/RHP)!**

Zeros have other implications that will will discuss

- They have the potential to “blank out” a signal at certain frequencies, making a system “blind” to it**
- Blanking out a crucial signal (e.g., a sensor reading) at the wrong point can have bad consequences**



Example





Partial Fractions

Consider a system of the following general form:

$$\mathbf{X(s)} = \frac{\mathbf{N(s)}}{\mathbf{D(s)}} = \frac{b_0 + b_1s + \dots + b_ms^m}{a_0 + a_1s + \dots + a_ns^n}$$

If the numerator and denominator polynomials are factored, we get an expression of the following form:

$$\mathbf{X(s)} = \frac{\mathbf{N(s)}}{\mathbf{D(s)}} = C \frac{(s - z_1)(s - z_2) \dots (s - z_m)}{(s - p_1)(s - p_2) \dots (s - p_n)}$$



Partial Fractions

$$\mathbf{X(s)} = \frac{\mathbf{N(s)}}{\mathbf{D(s)}} = C \frac{(s - z_1)(s - z_2) \dots (s - z_m)}{(s - p_1)(s - p_2) \dots (s - p_n)}$$

- **The number of zeros (m) and poles (n) is equal to the order of the numerator and denominator polynomials, respectively.**
- **Some of the poles and/or zeros may be complex**



Partial Fractions

$$\mathbf{X(s)} = \frac{\mathbf{N(s)}}{\mathbf{D(s)}} = C \frac{(s - z_1)(s - z_2) \dots (s - z_m)}{(s - p_1)(s - p_2) \dots (s - p_n)}$$

The orders of the numerator (m) and denominator (n) polynomials are important

$m < n$: Strictly proper

MOST CONVENIENT

$m = n$: Proper

$m > n$: Improper

Let's work through some examples to get the hang of it



Example 1

$$\mathbf{X}(s) = \frac{8}{(s+2)^2 + 1}$$

$$\Rightarrow \mathcal{L}^{-1} \left[\frac{8}{(s+2)^2 + 1} \right] = 8e^{-2t} \sin(t)u(t)$$

- We saw this example before
- $m = 0, n = 2$, so $m < n$, and it is strictly proper
- Very easy to invert



Example 2

$$\mathbf{X}(s) = \frac{\boxed{2s^2} + 8s + 6}{s^2 + 2s + 1} \quad m = n = 2 \Rightarrow \text{Proper}$$

**We need to rewrite this to get a strictly proper form,
by long division**

$$s^2 + 2s + 1 \overline{) \begin{array}{r} 2s^2 + 8s + 6 \\ \underline{2s^2 + 4s + 2} \\ 4s + 4 \end{array}} \Rightarrow \mathbf{X}(s) = \frac{\boxed{4(s+1)}}{s^2 + 2s + 1} + 2$$

STRICTLY PROPER



Example 3

$$\mathbf{X}(s) = \frac{6s^3 + 4s^2 + 8s + 6}{s^2 + 2s + 1} \quad m > n \Rightarrow \text{Improper}$$

$$\begin{array}{r} 6s \\ s^2 + 2s + 1 \overline{) 6s^3 + 4s^2 + 8s + 6} \\ \underline{6s^3 + 12s^2 + 6s} \\ -8s^2 + 2s + 6 \end{array}$$

$$\Rightarrow \mathbf{X}(s) = \frac{-8s^2 + 2s + 6}{s^2 + 2s + 1} + 6s$$

**PROPER, BUT NOT STRICTLY SO
(needs more work!)**



Example 3

$$X(s) = \frac{6s^3 + 4s^2 + 8s + 6}{s^2 + 2s + 1} \quad m > n \Rightarrow \text{Improper}$$

$$\begin{array}{r} 6s - 8 \\ s^2 + 2s + 1 \overline{) 6s^3 + 4s^2 + 8s + 6} \\ \underline{6s^3 + 12s^2 + 6s} \\ -8s^2 + 2s + 6 \\ \underline{-8s^2 - 16s - 8} \\ 18s + 14 \end{array} \quad \leftarrow \text{ORDER} < 2, \text{ so STOP}$$

$$\Rightarrow X(s) = \frac{18s + 14}{s^2 + 2s + 1} + 6s - 8$$

STRICTLY PROPER



What Next?

Once we have “cleaned up” $\mathbf{X}(s)$ and expressed it in terms of strictly proper functions, the next step is to expand the fraction. For example,

$$\begin{aligned}\mathbf{X}(s) &= \frac{s^2 - 4s + 3}{s(s+1)(s+3)} \\ &= \frac{A_1}{s} + \frac{A_2}{(s+1)} + \frac{A_3}{(s+3)}\end{aligned}$$

If we can figure out A_1 , A_2 , and A_3 , we will be all set to compute the inverse L.T.



The Residue Method

$$\mathbf{X(s)} = \frac{s^2 - 4s + 3}{s(s+1)(s+3)} = \frac{A_1}{s} + \frac{A_2}{(s+1)} + \frac{A_3}{(s+3)}$$

$$s\mathbf{X(s)} = A_1 + \frac{sA_2}{(s+1)} + \frac{sA_3}{(s+3)}$$

If we set $s = 0$, we get:

$$\mathbf{X(s)} = \frac{s^2 - 4s + 3}{s(s+1)(s+3)}$$

$$s\mathbf{X(s)} = A_1 = \left. \frac{s^2 - 4s + 3}{(s+1)(s+3)} \right|_{s=0} = 1$$



Partial Fractions (Cont.)

This was a particularly convenient case because the roots of the denominator were all real and distinct, so the poles were real and distinct

This does not always happen, so we need to learn how to deal with situations when either

- some of the poles are repeated, and/or**
- when they are complex**



Example

$$X(s) = \frac{s^2 + 3s + 3}{(s + 2)(s + 3)^3}$$

REPEATED REAL POLE

Partial Fraction Expansion:

$$X(s) = \frac{A}{(s + 2)} + \frac{B_1}{(s + 3)} + \frac{B_2}{(s + 3)^2} + \frac{B_3}{(s + 3)^3}$$



Example

$$X(s) = \frac{A}{(s+2)} + \underbrace{\frac{B_1}{(s+3)} + \frac{B_2}{(s+3)^2} + \frac{B_3}{(s+3)^3}}$$

EASY CASES:

$$A = (s+2)X(s)\big|_{s=-2}$$

$$B_3 = (s+3)^3 X(s)\big|_{s=-3}$$

The other two coefficients need a bit more work



Example

$$\mathbf{X}(s) = \frac{A}{(s+2)} + \underbrace{\frac{B_1}{(s+3)} + \frac{B_2}{(s+3)^2} + \frac{B_3}{(s+3)^3}}_{\text{partial fractions}}$$

$$(s+3)^3 \mathbf{X}(s) = \frac{A(s+3)^3}{(s+2)} + B_1(s+3)^2 + \underline{B_2(s+3)} + B_3$$

We can “isolate” B_2 by differentiation!

$$\left. \frac{d}{ds} (s+3)^3 \mathbf{X}(s) \right|_{s=-3} = B_2$$



Example

$$\mathbf{X}(s) = \frac{A}{(s+2)} + \underbrace{\frac{B_1}{(s+3)} + \frac{B_2}{(s+3)^2} + \frac{B_3}{(s+3)^3}}$$

$$(s+3)^3 \mathbf{X}(s) = \frac{A(s+3)^3}{(s+2)} + \underline{B_1(s+3)^2} + B_2(s+3) + B_3$$

Isolating B_1 will need differentiation twice!

$$\left. \frac{1}{2} \frac{d^2}{ds^2} (s+3)^3 \mathbf{X}(s) \right|_{s=-3} = B_1$$



Complex Poles

Complex poles can be distinct or repeated. Let's consider these cases individually, again through examples.

Example: Distinct Complex poles

$$\mathbf{X(s)} = \frac{4s + 1}{(s + 1)(s + 2 - j3)(s + 2 + j3)}$$

The fact that the poles are complex conjugates is not an accident.

– This is true whenever the coefficients of the polynomial are real



Distinct Complex Poles

$$\begin{aligned}\mathbf{X}(s) &= \frac{4s + 1}{(s + 1)(s + 2 - j3)(s + 2 + j3)} \\ &= \frac{A}{(s + 1)} + \frac{B_1}{(s + 2 - j3)} + \frac{B_2}{(s + 2 + j3)}\end{aligned}$$

Easy Case

$$B_1 = (s + 2 - j3)\mathbf{X}(s)\big|_{s=-2+j3}$$

$$B_2 = (s + 2 + j3)\mathbf{X}(s)\big|_{s=-2-j3}$$

Observe that: $B_2 = B_1^*$ This is true in general



Repeated Complex Poles

They occur as repeated pairs of complex conjugates.

For example:

$$\begin{aligned} \mathbf{X}(s) &= \frac{108(s^2 + 2)}{(s + 2)(s + 5 + j3)^2 (s + 5 - j3)^2} \\ &= \frac{A}{(s + 2)} + \frac{B_1}{(s + 5 + j3)} + \frac{B_2}{(s + 5 + j3)^2} \\ &\quad + \frac{B_3}{(s + 5 - j3)} + \frac{B_4}{(s + 5 - j3)^2} \end{aligned}$$

As before, we use the differentiation method to isolate B_2 and B_4



Activity

Calculate the inverse Laplace transform $x(t)$ for the following signal:

$$X(s) = e^{-2s} \frac{s^2 + s + 3}{s^3 + 2s^2 + s + 2} = e^{-2s} Y(s)$$

Step 1: Factor the denominator (it is easier than it looks)

Step 2: Do the partial fraction expansion for $Y(s)$

Step 3: Find the inverse Laplace Transform $y(t)$

Step 4: Apply the delay of 2 to the answer from Step 3

Correct answer: $x(t) = \sin(t - 2)u(t - 2) + e^{-2(t-2)}u(t - 2)$



Summary

Today, we learned to expand a Laplace Transform that looks like a ratio of polynomials into a partial fraction series

$$\mathbf{X(s)} = \frac{\mathbf{N(s)}}{\mathbf{D(s)}} = \frac{b_0 + b_1s + \dots + b_ms^m}{a_0 + a_1s + \dots + a_ns^n}$$

$m < n$: Strictly proper

$m = n$: Proper

$m > n$: Improper

This is very helpful, since it allows us to figure out the inverse Laplace transform by simple Table lookup

We learned to interpret poles and zeros



For the Next Class

The Canvas site provides a bunch of partial fraction problems with answers for your practice

Work through the textbook examples in Section 3.5

For the next class, please study Sections 3.6 – 3.8

How to Study:

1. Read the assigned sections in order to understand the basic concepts, and aim to memorize the concepts
2. Attempt the in-chapter exercises and practice until you can do them fully on your own without notes, like in a test
3. Write down your questions and bring them to class